MAJOR/MINOR PROJECT SYNOPSIS

ON

Privacy-Preserving Gaze and Affective State Analysis for Secure Digital Environments

AMITY SCHOOL OF ENGINERING AND TECHNOLOGY

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1. ABSTRACT

The transition to remote academic assessments has exposed critical vulnerabilities in current proctoring systems, specifically regarding data privacy and the high computational cost of cloud-based monitoring. Sentin-Edge AI is a decentralized solution designed to perform real-time monitoring of user gaze and affective states (stress levels) entirely on the local device. By utilizing Edge-AI, the system ensures that sensitive visual data remains on the host machine, mitigating privacy risks. The integration of emotional intelligence through facial micro-expression analysis provides a nuanced understanding of user behavior, reducing false positives in integrity monitoring.

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2. INTRODUCTION

Traditional proctoring tools often rely on continuous video streaming to centralized servers, which leads to significant bandwidth consumption and potential data breaches. Furthermore, these systems are typically binary, flagging any deviation in gaze as a violation without considering the user's psychological context (e.g., concentration or confusion).

Sentin-Edge AI addresses these challenges by processing video frames in-memory to extract mathematical landmarks, which are then analyzed locally. The project introduces an "Affective Layer" that correlates physiological stress markers with gaze patterns. This research aims to prove that privacy-compliant, context-aware monitoring is achievable on standard consumer hardware without compromising accuracy.

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3. METHODOLOGY

The project employs a Multi-Task Learning (MTL) approach to execute simultaneous behavioral analysis streams.

- Landmark Extraction Pipeline: Using MediaPipe Face Mesh, 478 3D landmarks are localized. Specific attention is given to the iris and periorbital regions to calculate precise gaze vectors.
- Dual-Head Classification: * Gaze Head: A lightweight classifier (CNN or Random Forest) maps ocular coordinates into zones: *Center, Left, Right, and Off-Screen.*
    - Affective Head: A Temporal Convolutional Network (TCN) monitors the frequency and intensity of landmark shifts in the brow and mouth regions to identify "Stress" or "Neutral" states.
- Weighted Logic Engine: The system applies a weighted score to events. For instance, an "Off-Screen" gaze combined with a "High Stress" spike triggers a high-probability alert, whereas the same gaze with a "Neutral" state is categorized as a cognitive pause (thinking).
- Optimization Research: The methodology includes a comparative study of the Full Model vs. an 8-bit Quantized Model to measure the trade-off between inference speed and the sensitivity of micro-expression detection.

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4. TOOLS AND TECHNOLOGIES

The implementation will utilize an open-source, high-performance stack optimized for edge deployment:

- Programming Language: Python 3.9+
- Computer Vision: OpenCV (Frame processing) and MediaPipe (Feature extraction).
- Deep Learning Framework: TensorFlow / Keras for model development.
- Edge Deployment: TensorFlow Lite or ONNX Runtime for local CPU optimization.
- Statistical Analysis: Scikit-Learn for performance benchmarking.
- Hardware: Standard laptop equipped with a 720p/1080p Integrated Webcam.

5. PROJECT PLAN

The project is structured into four comprehensive phases, distributed over a 16-week academic semester to ensure robust development and rigorous testing:

- **Phase 1: Research, Environment & Preprocessing (Weeks 1-4):**
  - **Literature review of current Edge-AI and Gaze tracking methodologies.**
  - **Setting up the Python development environment and integrating MediaPipe for real-time 3D landmark extraction.**
  - **Developing the geometric logic for initial ocular movement detection and data logging.**
- **Phase 2: Data Acquisition & Model Training (Weeks 5-8):**
  - **Preprocessing landmark-only data from standard datasets (MPIIGaze and FER-2013).**
  - **Designing and training the Gaze Classification Head and the Affective State Head (Temporal Convolutional Network).**
  - **Validating models against test sets to ensure high precision in identifying micro-expressions.**
- **Phase 3: System Integration & Edge Optimization (Weeks 9-12):**
  - **Merging the dual-head models into a unified Multi-Task Learning (MTL) pipeline.**
  - **Implementing 8-bit Integer Quantization and converting the architecture into TensorFlow Lite format for local CPU inference.**
  - **Developing the "Integrity Engine" logic to correlate stress spikes with gaze deviation.**
- **Phase 4: Benchmarking, Testing & Documentation (Weeks 13-16):**
  - **Conducting comparative performance analysis between the Full and Quantized models (Latency vs. Accuracy).**
  - **Finalizing the user interface (UI) for the local monitoring dashboard.**
  - **Compiling the final technical research report and preparing the project presentation for the evaluation committee.**